JAYDIP NAKARANI

Senior Machine Learning Engineer

Surat, Gujarat, India • jaydipnakrani888@gmail.com • +91 78028 30436

PROFILE SUMMARY

Machine Learning Engineer with 8+ years of experience designing, training, and deploying production ML systems across Computer Vision, NLP, and Deep Learning. Strong track record of building end-to-end ML pipelines — from data ingestion and ETL, to model training, evaluation, packaging, and scalable deployment — using PyTorch, Kubeflow Pipelines, Docker, and AWS. Solid foundation in MLOps, reproducible training workflows, and agentic AI systems built on top of large language models.

CORE SKILLS

ML / DL Frameworks: PyTorch, TensorFlow, Scikit-Learn, Hugging Face Transformers, XGBoost, LightGBM

Computer Vision: YOLOv8, Detectron2, OpenCV, StyleGAN, Image Annotation (Roboflow), OCR, OCR fine-tuning

NLP & LLMs: spaCy, BERT / Transformers, Fine-tuning, Prompt Engineering, RAG, OpenAI, Claude, LangChain, LangGraph, LlamaIndex

MLOps & ML Platform: Kubeflow Pipelines (kfp), SageMaker-style training pipelines, Model Registry, CI/CD for ML, experiment tracking

Data Engineering: ETL Pipelines, Apache Spark, PyMySQL, StarRocks, PostgreSQL, MongoDB, Feature Engineering, Data Lakes

Python Ecosystem: Python 3.x, Poetry, uv, Pydantic, FastAPI (model serving), Flask

Testing & Code Quality: pytest (pytest-cov, pytest-mock, pytest-xdist), tox, Black, isort, mypy

Cloud & DevOps: AWS (EC2, S3, Lambda, SageMaker, RDS), GCP, Docker, Kubernetes, Git / GitLab

Other: TDD, API Documentation (Swagger), Performance Optimization, Algorithms & Data Structures

PROFESSIONAL EXPERIENCE

Sr. Machine Learning Engineer

BitCoding Solutions *Feb 2022 – Apr 2026*

- Led end-to-end delivery of production ML systems across Computer Vision, NLP, multimodal, and agentic AI domains for enterprise clients.
- Architected model training pipelines covering data preparation, annotation, fine-tuning, evaluation, packaging, and containerised deployment on AWS.
- Designed multi-agent AI orchestration frameworks on top of LLMs using LangGraph, LangChain, and LlamaIndex with RAG-based knowledge retrieval.
- Built automated training / evaluation / retraining loops to monitor model drift and continuously improve underperforming models.
- Mentored engineers on ML engineering standards, reproducible experiments, and code quality (Black, isort, mypy, pytest).

Machine Learning Engineer

GlobaliaSoft LLP *Dec 2020 – Jan 2022*

- Built an internal ML data platform powering centralised ETL, feature engineering, and model training workflows for data-heavy client products.
- Implemented Kubeflow Pipelines (kfp) for reproducible training, hyperparameter tuning, and deployment stages, integrating Apache Spark for large-scale feature computation.

- Set coding standards for the ML codebase (Poetry, Black, isort, mypy, pytest) and mentored junior developers on ML engineering best practices.

Freelance Python / Data Engineer

Self-employed *Jul 2017 – Nov 2020*

- Delivered large-scale web scraping and ETL solutions across real-estate, finance, travel, e-commerce, and social-media domains.
- Designed scraping pipelines with proxy rotation on AWS Lambda, followed by ETL jobs that cleaned and structured millions of records for downstream analytics and ML feature engineering.
- Practiced TDD, managed small projects directly with clients, and delivered structured data in CSV, JSON, SQL, and cloud storage formats.

KEY PROJECTS

Vektr – Multi-Agent Orchestration System *Apr 2025 – Present*

Python, LangGraph, LangChain, OpenAI, Claude, LlamaIndex, Neo4j, PyTorch, spaCy, FastAPI, React, MCP

- Designed a multi-agent automation framework using LangGraph as the meta-layer orchestrator, with specialised onboarding, design, build, and training agents.
- Integrated LlamaIndex for RAG-based retrieval and Neo4j for graph-structured knowledge and relationships accessed by agents at runtime.
- Built an automated training / testing / retraining pipeline using PyTorch and spaCy preprocessing; agents with low performance are flagged and retrained automatically.
- Developed a TypeScript/React dashboard for monitoring agent-level KPIs and department-level performance.

KYC Document Verification System *Oct 2024 – Mar 2025*

Python, PyTorch, YOLOv8, Roboflow, FastAPI, Docker, AWS (EC2, S3, RDS), PostgreSQL, pytest

- Fine-tuned YOLOv8 for Aadhaar and PAN card detection, field localisation, and structured data extraction (Name, DOB, Aadhaar/PAN number, signature, address).
- Curated and annotated training datasets in Roboflow; built an evaluation pipeline tracking precision, recall, and field-level accuracy across test splits.
- Packaged the model as a Dockerised inference service and deployed on AWS EC2 with S3 for document storage and RDS for metadata.
- Wrote pytest regression suites covering inference endpoints, edge-case documents, and model-output schema validation.

Videoreach – Medical Protocol to Voice + Video Generation *Feb 2024 – Sep 2024*

Python, Claude API, Pydantic AI, LlamaParse, Eleven Labs (voice cloning), FastAPI, Logfire

- Built a multi-agent NLP pipeline extracting structured medical protocol text (endoscopy, colonoscopy, etc.) from unstructured clinical documents via LlamaParse.
- Implemented a validation layer to gate generation on extraction confidence, and a script-generation agent converting protocols into episodic patient instructions.
- Integrated Eleven Labs for doctor-voice cloning and a video-sync agent producing the final instructional video output.

Intelligent Invoice Processing System *Aug 2023 – Jan 2024*

Python, Claude 3.5, OpenCV, PDF parsing, QR decoding, Pydantic, FastAPI, Docker, pytest

- Built an AI-driven extraction pipeline for digital and scanned invoices, combining OCR, layout heuristics, and LLM-assisted field extraction.
- Implemented QR decoding for government-linked verification and a validation agent cross-checking extracted fields against official portals.

- Delivered structured JSON output with full audit logs and monitoring for every extraction and validation step, achieving high accuracy across diverse invoice formats.

Real-Time Hyper-Realistic Talking Avatar *May 2023 – Jul 2023*

Python, PyTorch, StyleGAN, SMPL, MuseV, MuseTalk, MusePose, OpenAI STT/TTS, Docker, AWS (ECS), FastAPI

- Designed an end-to-end avatar system integrating MuseV (visual rendering), MuseTalk (TTS + lip-sync), and MusePose (gesture animation) for real-time conversational avatars.
- Integrated StyleGAN for realistic facial expressions and CMU MoCap data for natural motion patterns.
- Replaced initial OpenAI STT/TTS with custom-trained models to improve latency and tailor voice characteristics to client requirements.
- Containerised and deployed on AWS ECS, tuning inference throughput for real-time use.

Vendor Product Data Management System *Jan 2023 – Apr 2023*

Python, Scikit-Learn, Ensemble ML, NLP, spaCy, AWS Textract, FastAPI, AWS, SQL

- Designed a header-classification model for structured data extraction from Excel/CSV catalogs and an NLP-based parser for unstructured PDF catalogs.
- Built an ensemble ML classifier tagging products with business-relevant attributes, improving catalogue search and recommendation quality.
- Implemented a validation layer and real-time price / attribute update pipeline maintaining database accuracy across frequent vendor changes.

Car Damage Detection & Signature / QR Extraction *Sep 2022 – Dec 2022*

Python, PyTorch, Detectron2, OpenCV, pyzbar, FastAPI, Docker, PostgreSQL, AWS

- Trained a Detectron2 instance-segmentation model for multi-class car damage detection; annotated training data in Roboflow and benchmarked on an independent test set.
- Developed OpenCV-based signature extraction robust to diverse document styles and noise levels.
- Deployed trained models as Dockerised FastAPI endpoints with PostgreSQL storing results and metadata.

Document Layout Understanding & Custom OCR Pipeline *Feb 2022 – Aug 2022*

Python, PyTorch, LayoutLMv3, Hugging Face Transformers, Tesseract OCR, OpenCV, FastAPI, Docker, AWS (EC2, S3), pytest

- Fine-tuned LayoutLMv3 for document key-value extraction across forms, contracts, and structured business documents.
- Built an end-to-end data preparation pipeline covering image normalisation, OCR token alignment, bounding-box annotation, and train / validation splits.
- Trained a custom OCR head on noisy and low-resolution scans, improving extraction accuracy on legacy documents over off-the-shelf OCR baselines.
- Packaged the pipeline as a Dockerised FastAPI inference service deployed on AWS EC2, with S3 for document upload and result storage.
- Wrote pytest suites covering layout-prediction outputs, OCR accuracy benchmarks, and end-to-end inference contracts.

NLP & Document Classification Microservices *Aug 2021 – Jan 2022*

Python, spaCy, PyTorch, Hugging Face Transformers, Scikit-Learn, FastAPI, Docker, AWS (EC2, S3), PostgreSQL, pytest

- Built NLP microservices for text extraction, named-entity recognition, and rule-based + ML document classification across client data products.
- Fine-tuned spaCy and lightweight Transformer models for domain-specific entity extraction and category tagging on noisy real-world text.
- Packaged each model as a FastAPI inference service with Docker, deployed on AWS EC2; integrated with the internal ML data platform for training data flow.
- Wrote pytest-based unit and integration tests covering preprocessing, model outputs, and API contracts to guarantee reproducible deployments.

Enterprise ML Data Platform & ETL Pipeline *Dec 2020 – Jul 2021*

Python, Kubeflow Pipelines (kfp), Apache Spark, PyMySQL, StarRocks, Docker, Poetry, pytest, mypy, Black

- Built an internal ML data platform powering training data preparation, feature engineering, and reproducible model training workflows.
- Authored Kubeflow Pipelines (kfp) components covering ingestion, transformation, training, and deployment — each step containerised with Docker for isolation and reproducibility.
- Implemented Apache Spark ETL jobs transforming raw catalogue and transactional data into feature tables consumed by downstream models.
- Integrated StarRocks (accessed via PyMySQL) as the analytical store backing large-scale feature retrieval and offline evaluation.
- Enforced code quality across the platform with pytest-cov, pytest-xdist, mypy, Black, and isort, and set up tox-based CI for multi-environment testing.

Banking Data Automation & Anomaly Detection *May 2020 – Nov 2020*

Python, Selenium, Scikit-Learn, Pandas, NumPy, PostgreSQL, AWS (EC2, S3), Cron, Flask

- Built Selenium-based automation pipelines extracting structured transaction data from secured banking portals across multiple client accounts.
- Designed a normalisation layer converting heterogeneous bank statement formats into a unified transaction schema stored in PostgreSQL.
- Trained Scikit-Learn classifiers for transaction categorisation and an Isolation Forest model for anomaly detection on suspicious transactions.
- Exposed model outputs through a Flask service and scheduled extraction + scoring jobs via cron on AWS EC2.

ETL & Data Delivery Pipelines for Analytics Clients *Apr 2019 – Apr 2020*

Python, Pandas, NumPy, SQL, AWS S3, Oracle, Dropbox API, JSON / CSV / XLS / XML, Cron

- Designed end-to-end ETL pipelines transforming scraped and client-provided data into structured feeds for analytics and BI tools.
- Implemented schema validation, deduplication, and incremental update logic ensuring data integrity across daily and real-time delivery cycles.
- Automated data delivery to AWS S3, Dropbox, Oracle, and SQL warehouses, supporting CSV, JSON, XML, XLS, and SQL output contracts.
- Built monitoring and retry mechanisms for long-running jobs and integrated logging for downstream consumer visibility.

Large-Scale Web Scraping Platform *Jul 2017 – Mar 2019*

Python, Scrapy, Selenium, BeautifulSoup, LXML, Requests, AWS Lambda, S3, Proxy Management, Cloudflare bypass, Pandas

- Designed a master–slave scraping architecture on AWS Lambda with proxy rotation to bypass Cloudflare and collect millions of records reliably.
- Delivered scraping projects across multiple verticals — real estate, e-commerce (Amazon, eBay, Alibaba), social media (Twitter, YouTube, Reddit, Instagram), search (Google Maps, Bing), and reviews (Yelp).
- Implemented robust CAPTCHA handling, request throttling, and headless browser automation for JavaScript-heavy and protected sources.
- Structured outputs into CSV, JSON, XML, and SQL formats and pushed them to AWS S3 and downstream client storage.

EDUCATION

B. Tech – Computer Science & Engineering

Parul University, Vadodara | *Aug 2016 – Apr 2020*

CERTIFICATIONS

- Amazon Web Services (AWS): Cloud Essentials, Architecting, File Storage, Serverless, Compute, Migration Foundations
- IBM Watson and Machine Learning: Technical Essentials, Machine Learning Essentials
- Remote Work Professional Certification (RWPC™)

ADDITIONAL DETAILS

Languages: English, Hindi, Gujarati

Strengths: Quick learner, strong communicator, analytical thinker, problem-solver, team player, self-motivated.